

Chapter 4

Production theory

In this chapter we study production theory in a commodity space. First, we develop the mathematical representation of production units, which we refer simply as firms. Second, we characterize the “behavior” of firms whose objective is the maximization of profits.

4.1 Representation of firms

We study an L -commodity world. Here, a firm is an economic entity that transforms commodities. We represent a firm by means of the collection of bundles in the commodity space that are “technically feasible” for it.

Example 13. Assume that $L = \{\text{oranges, orange juice}\}$. We denote the generic amount of oranges, in pounds, by z_1 and the generic amount of orange juice, in liters, by y_1 . A given firm is able to produce from each pound of oranges, half a pound of juice, and cannot produce any oranges from orange juice. Moreover, the firm can dispose of any excess of orange juice. The mathematical representation of this firm is the set

$$Y \equiv \left\{ (-z_1, y_1) \in \mathbb{R}^2 : z_1 \geq 0, y_1 \leq \frac{1}{2}z_1 \right\}.$$

Definition 97. A **production set** is a subset of $\mathbb{R}^L$. The generic production set is Y .

We refer to the elements of Y as **feasible vectors** (for Y). For each $y \in Y$, and each $l \in \{1, \dots, L\}$, commodity l is an input at y whenever $y_l < 0$; symmetrically, it is an output whenever $y_l > 0$.

The following two examples will guide our study of production sets.

Example 14 (Transformation Function). We define a production set by means of a function that tells us which vectors are feasible. No restriction is imposed on whether a commodity is an input or an output.

Let $F : \mathbb{R}^L \rightarrow \mathbb{R}$ be a function. The production set associated with F is

$$Y^F \equiv \{y \in \mathbb{R}^L : F(y) \leq 0\}.$$

We refer to F as a **transformation function** and to the set

$$\partial F \equiv \{y \in \mathbb{R}^L : F(y) = 0\},$$

as the **transformation frontier** for F .

We define the **Marginal Rate of Transformation (MRT) of commodity l for commodity k at $y \in \partial F$** as the ratio:

$$MRT_{lk}(y) \equiv \frac{\frac{\partial F}{\partial y_l}(y)}{\frac{\partial F}{\partial y_k}(y)}.$$

The MRT of commodity l for commodity k at y is the absolute value of the slope of the transformation frontier on the l - k “slide through y ” of the consumption space.

Example 15 (Single-output model). We divide the commodity space into inputs and outputs. We define a production set by means of a function that tells us the amount of output that can be produced with a given input vector. For simplicity we restrict to the single-output case.

Let $f : \mathbb{R}_+^{L-1} \rightarrow \mathbb{R}_+$ be a function. The production set associated with f is

$$Y^f \equiv \{(-z, q) \in \mathbb{R}_+^L : z \equiv (z_1, \dots, z_{L-1}) \in \mathbb{R}_+^{L-1} \text{ and } q - f(z_1, \dots, z_{L-1}) \leq 0\}.$$

We refer to f as a **production function** and to the set

$$\partial f \equiv \{(-z, q) \in \mathbb{R}_+^L : z \in \mathbb{R}_+^{L-1} \text{ and } q - f(z_1, \dots, z_{L-1}) = 0\},$$

as the **production frontier**.

We define the **Marginal Rate of Technical Substitution (MRTS) of input l for input k at $z \in \mathbb{R}_+^{L-1}$** as the ratio:

$$MRTS_{lk}(z) \equiv \frac{\frac{\partial f}{\partial z_l}(z)}{\frac{\partial f}{\partial z_k}(z)}.$$

The single-output model admits an alternative representation of the production set by means of iso-quant. The MRTS of input l for input k at z is the absolute value the slope of the iso-quant at z .

4.2 Properties of production sets

Let $Y \subseteq \mathbb{R}^L$ be a production set. Then Y is (or satisfies):

1. **Non-empty:** if $Y \neq \emptyset$.
2. **Closed:** if Y is closed in $\mathbb{R}^L$.
3. **No-free-lunch:** for each $y \in Y$, if $y \geq 0$, then $y = 0$. Alternatively, $Y \cap \mathbb{R}_+^L \subseteq \{0\}$.
4. **Possibility of inaction:** $0 \in Y$.
5. **Free-disposal:** if $y \in Y$, then for each $x \leq y$, $x \in Y$.
6. **Irreversible:** $y \in Y$ and $y \neq 0$, then $-y \notin Y$.
7. **Non-increasing returns to scale (NI):** for each $y \in Y$ and each $\alpha \in [0, 1]$, $\alpha y \in Y$. (Does it imply convexity?)
8. **Non-decreasing returns to scale (ND):** for each $y \in Y$, and each $\alpha \in [1, +\infty)$, $\alpha y \in Y$. (Does it imply convexity? Is it independent of non-increasing returns to scale?)
9. **Constant returns to scale (CR):** satisfies both NI and ND returns to scale. (Does it imply convexity? What about in conjunction with free disposal?)
10. **Additivity:** for each pair $\{y, y'\} \subseteq Y$, $y + y' \in Y$.
11. **Convexity:** Y is convex.

Proposition 48. Let $Y \subseteq \mathbb{R}^L$.

1. Suppose that $L = 2$ and $Y \subseteq \mathbb{R}_- \times \mathbb{R}$ (single-output model). If Y satisfies free-disposal and CR of scale, then Y is convex.
2. Suppose that $L = 3$ and $Y \subseteq \mathbb{R}_-^2 \times \mathbb{R}$ (single-output model). Then, free-disposal and CR of scale, may not imply that Y is convex.

Proposition 49. Let $Y \subseteq \mathbb{R}^L$. If Y is additive and satisfies NI returns to scale, then Y is a convex cone, i.e., for each pair $\{y, y'\} \subseteq Y$ and each pair $\{\alpha, \beta\} \in \mathbb{R}_+$, $\alpha y + \beta y' \in Y$.

Proof. Let $\{y, y'\} \subseteq Y$ and $\{\alpha, \beta\} \in \mathbb{R}_+$. If $\alpha \in [0, 1]$ then NI returns to scale imply that $\alpha y \in Y$. If $\alpha \geq 1$, then let $n \in \mathbb{N}$ be such that $\frac{\alpha}{n} < 1$. Then, by NI returns to scale, $\frac{\alpha}{n} y \in Y$. By additivity, $\alpha y = \frac{\alpha}{n} y + \cdots + \frac{\alpha}{n} y \in Y$. The symmetric argument shows that $\beta y' \in Y$. By additivity, $\alpha y + \beta y' \in Y$. $\square$

Corollary 11. Additivity and NI returns to scale imply CR returns to scale and convexity.

4.3 Profit maximization

In this section we characterize the behavior of a price-taker profit maximizing firm.

Consider a firm represented by production set Y . Assume that prices are $p \in \mathbb{R}_+^L$. The profit of firm Y when producing at a feasible vector $y \in Y$ is

$$p \cdot y \equiv \sum_{l \in L} p_l y_l.$$

A price-taker profit maximizing firm that observes prices p solves the **Profit Maximization Problem**

$$\max_{y \in Y} p \cdot y.$$

The solution of this problem and the maximum profit attained by the firm define the firm's supply correspondence and profit function.

Definition 98. Let Y be a production set. The **supply correspondence associated with Y** is defined by

$$\begin{aligned} y^Y : \mathbb{R}_+^L &\rightarrow \mathbb{R}^L \\ p &\mapsto y^Y(p) \equiv \arg \max_{y \in Y} p \cdot y. \end{aligned}$$

Definition 99. Let Y be a production set. The **profit function associated with Y** is defined by:

$$\begin{aligned} \Pi^Y : \mathbb{R}_+^L &\rightarrow \mathbb{R}^L \\ p &\mapsto \Pi^Y(p) \equiv \max_{y \in Y} p \cdot y. \end{aligned}$$

Alternatively, $\Pi^Y(p) \equiv p \cdot y^*$ for some $y^* \in y^Y(p)$.

4.3.1 Profit maximization: Transformation Function model

If production set Y is described by means of a Transformation function F , the firm's profit maximization problem becomes

$$\max_{\{y \in \mathbb{R}^L : F(y) \leq 0\}} p \cdot y.$$

We simplify notation and denote y^F and Π^F the supply correspondence and profit function associated with Y^F .

Proposition 50. Let F be a differentiable transformation function. For each $p \in \mathbb{R}_{++}^L$ and each pair $\{l, k\} \subseteq \{1, \dots, L\}$,

$$\frac{p_l}{p_k} = MRT_{lk}(y^F(p))$$

4.3.2 Profit maximization: Production Function model

In the production function single-output model we denote input prices by $w \equiv (w_1, \dots, w_{L-1}) \in \mathbb{R}_+^{L-1}$ and output price by $p \in \mathbb{R}_+$.

If production set Y is described by means of a Production function f , the firm's profit maximization problem becomes:

$$\max_{z \in \mathbb{R}_+^{L-1}} pf(q) - w \cdot z.$$

We simplify notation and denote y^f and Π^f the supply correspondence and profit functions associated with Y^f .

Proposition 51. Let f be a differentiable production function and Y its associated production set. For each $w \in \mathbb{R}_{++}^{L-1}$, each $p \in \mathbb{R}_{++}$, and each pair $\{l, k\} \subseteq \{1, \dots, L\}$,

$$\frac{w_l}{w_k} = MRTS_{lk}(y^f(p))$$

4.3.3 Profit maximization: general model

The following are properties of a supply correspondence and a profit function.

Proposition 52. Assume that Y is non-empty, closed, and satisfies free disposal. Then:

1. Π^Y is homogeneous of degree one.

2. Π^Y is convex.
3. If Y is convex, then it can be recovered from Π^Y as follows:

$$Y \equiv \{y \in \mathbb{R}^L : \forall p \in \mathbb{R}_+^L, p \cdot y \leq \Pi^Y(p)\}.$$

4. y^Y is homogeneous of degree zero.
5. If Y is convex, then y^Y is convex-valued. If Y is strictly convex, i.e., for each pair $\{x, y\} \subseteq Y$ and each $\alpha \in (0, 1)$, $\alpha x + (1 - \alpha)y$ belongs to the interior of Y , then y^Y is single-valued.
6. (Hotteling's Lemma) Let $p \in \mathbb{R}_+^L$. If $|y^Y(p)| = 1$, then Π is differentiable for each $l \in \{1, \dots, L\}$,

$$\frac{\partial \Pi^Y}{\partial p_l}(p) = y_l^Y(p).$$

7. If y^Y is a differentiable function in a neighborhood of $p \in \mathbb{R}_+^L$, then

- (a) $Dy^Y(p)p = 0$,
- (b) $Dy^Y(p)$ is symmetric and positive semi-definite.

8. (Law of supply) For each pair $\{p, p'\} \subseteq \mathbb{R}_+^L$,

$$(p' - p) \cdot (y^Y(p') - y^Y(p)) \geq 0.$$

4.4 Problems

1. Suppose that f is the production function associated with a single-output technology, and let Y be the associated production set. Prove that Y satisfies Constant Returns of Scale if and only if f is homogeneous of degree one. Conclude that in the single-input single-output case CR implies that f is linear.
2. Consider the single-input case in which f is defined by: for each $z \in \mathbb{R}_+$,

$$f(z) \equiv \frac{1}{2}(\sqrt{z+1} - 1).$$

- (a) Graph Y
- (b) Graphically show that Y is closed and satisfies Non-increasing returns to scale.

- (c) Graphically show that Y is not additive and does not satisfy Non-decreasing returns to scale.
- (d) Imagine now that it is possible to add any number of production input-output combinations under technology Y (there is free entry to this market). Denote this new production set by AY . Formally,

$$AY \equiv \{y_1 + \cdots + y_n : n \in \mathbb{N}, \forall i = 1, \dots, n, y_i \in Y\}.$$

- i. Characterize and graph AY . Is there a function f_A such that $AY \equiv \{(-z, q) : z \geq 0, q \leq f_A(z)\}$?
 - ii. Is AY closed? Is it additive? Does it satisfy Non Increasing Returns to scale? Does it satisfy Non Decreasing Returns to scale?
3. Consider a production set $Y \subseteq \mathbb{R}^L$. Let π^Y be the profit function associated with set Y , i.e., for each $p \in \mathbb{R}_+^L$,

$$\pi^Y(p) \equiv \max_{y \in Y} p \cdot y.$$

- (a) Define free disposal and no free lunch. Draw a production set that satisfies these two properties. Illustrate the profit maximization for some price vector p .
- (b) Is π^Y concave, convex, or neither of them?
- (c) Suppose now that a firm with production set Y may face an uncertain market. The firm is an expected profit maximizer and is **risk neutral**. There are two markets in which the firm may participate.
 - Market 1: price is uncertain with some probability distribution.
 - Market 2: price is certain; it is exactly the expected price in Market 1.

Prove that the firm always finds participating in Market 1 as good as participating in Market 2.

4. Let C be a convex and closed set of $\mathbb{R}^L$ and $z \in \mathbb{R}^L$ a vector that is not in C , i.e., $z \in C^c$. Prove that there is a closest vector in C to z , that is, there is $x^* \in C$ such that for each $x \in C$, $\|z - x^*\| \leq \|z - x\|$. Hint: the function $f : \mathbb{R}^L \rightarrow \mathbb{R}$ defined by $x \in \mathbb{R}^L \mapsto f(x) \equiv \|z - x\|^2$ is continuous.

5. Let C be a convex and closed set of $\mathbb{R}^L$ and $z \in \mathbb{R}^L$ a vector that is not in C . Let $x^* \in C$. Prove that x^* is such that for each $x \in C$, $\|z - x^*\| \leq \|z - x\|$ if and only if for each $x \in C$,

$$(z - x^*) \cdot (x - x^*) \leq 0.$$

Moreover, there is a unique closest vector in C to z . Do it in two steps.

- (a) Suppose that $x^* \in C$ is such that for each $x \in C$,

$$(z - x^*) \cdot (x - x^*) \leq 0.$$

Prove that x^* is the unique closest element in C to z . Do it as follows: prove that the function f defined above is strictly convex. Let $x \in \mathbb{R}^L$ be such that $x \neq x^*$. Prove, using a Taylor's expansion of degree two, that $f(x) > f(x^*)$.

- (b) Assume that x^* is a closest element in C to z . Let $x \in C$. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $t \mapsto \varphi(t) \equiv \|z - (x^* + t(x - x^*))\|^2$. Prove that φ is differentiable and that $\varphi'(0) \geq 0$. Conclude that $(z - x^*) \cdot (x - x^*) \leq 0$.

6. (Basic Separation Theorem) Suppose that C is a convex and closed set of $\mathbb{R}^L$ and $z \in \mathbb{R}^L$ a vector that is not in C . Prove that there is a non-zero vector $p \in \mathbb{R}^L$ and a real number c such that $p \cdot z > c$ and for each $x \in C$, $p \cdot x < c$. Do it in three steps.

- Provide a graphical interpretation of the theorem when C is a production set and z is a non-feasible point at C .
- Let $x^* \in C$ be the closest point in C to z . Let $p \equiv z - x^*$. Prove that $p \cdot z > p \cdot x^*$.
- Let $c \in \mathbb{R}$ be such that $p \cdot x^* < c < p \cdot z$. Prove that p and c have the desired properties.

7. (Basic Supporting hyperplane Theorem) Let C be a convex set of $\mathbb{R}^L$ and z be a boundary point of C , i.e., $z \in \overline{C} \cap \overline{C^c}$. (Here if A is a set, $\overline{A}$ denotes the closure of A , i.e., the set of all limit points of A . In other words, the set of points, a such that there is a sequence in A that converges to a . For instance, the closure of $(0, 1)$ is $[0, 1]$.) Prove that there is a non-zero vector $p \in \mathbb{R}^L$ such that for each $x \in C$, $p \cdot x \leq p \cdot z$. Provide a graphical interpretation when C is a production set and z is a boundary point of C .

8. Let $Y \subseteq \mathbb{R}^L$ be a non-empty convex production set that satisfies free disposal. Let y be efficient at Y , i.e., there is no $y' \in Y$ such that $y' \succ y$. Prove that there is $p \geq 0$ such that y is profit maximizing at p .
9. Let $Y \subseteq \mathbb{R}^L$ be a non-empty, convex, closed production set that satisfies free disposal. Prove that

$$Y = \{y \in \mathbb{R}^L : \forall p \geq 0, p \cdot y \leq \pi^Y(p)\}.$$

Is it always true that

$$Y = \{y \in \mathbb{R}^L : \forall p \gg 0, p \cdot y \leq \pi^Y(p)\}?$$

Would your answer change if Y also satisfies no free-lunch?

4.5 Cost minimization

In the context of the Production function model we can unambiguously refer to inputs and hence keep track of a firm's costs. In this section we characterize a firm's profit maximization problem by means of its cost minimization problem. For simplicity, we will concentrate in the single-output case.

Consider a firm represented by a production function f . Imagine that the firm decides to produce q units of output and input prices are $w \in \mathbb{R}_+^L$. The firm's Cost Minimization Problem for q at w is:

$$\min_{\{z \in \mathbb{R}_+^L : f(z) \geq q\}} w \cdot z.$$

Definition 100. Let f be a production function. The **cost function associated with f** is defined by:

$$\begin{aligned} c^f : \quad \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ (w, q) &\mapsto c^f(w, q) \equiv \min_{\{z \in \mathbb{R}_+^L : f(z) \geq q\}} w \cdot z. \end{aligned}$$

Definition 101. Let f be a production function. The **conditional factor demand correspondence associated with f** is defined by:

$$\begin{aligned} z^f : \quad \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ (w, q) &\mapsto z^f(w, q) \equiv \arg \min_{\{z \in \mathbb{R}_+^L : f(z) \geq q\}} w \cdot z. \end{aligned}$$

Proposition 53. Let f be a production function, $w \in \mathbb{R}_+^{L-1}$ and $q \in \mathbb{R}_+$. Assume that $|z^f(w, q)| = 1$ and $z^f(w, q) \gg 0$. Then:

1. For each pair $\{l, k\} \subseteq \{1, \dots, L-1\}$,

$$\frac{w_l}{w_k} = MRTS_{lk}(z^f(w, q)).$$

2. If λ is the Lagrange Multiplier associated with the Cost Minimization Problem for the constraint $q - f(z) \leq 0$, then

$$\frac{\partial c}{\partial q}(w, q) = |\lambda|.$$

The following are properties of a cost function.

Proposition 54. Let f be a production function. Then:

1. c^f is homogeneous of degree one in w .
2. For each $q \in \mathbb{R}_+$, $c^f(\cdot, q)$ is a concave function.
3. If for each $q > 0$, $\{z \in \mathbb{R}_+^{L-1} : f(z) \geq q\}$ are convex sets, then

$$Y^f \equiv \{(-z, q) : z \geq 0, \forall w \gg 0, w \cdot z \geq c(w, q)\}.$$

4. z^f is homogeneous of degree zero in w .
5. If for each $q \in \mathbb{R}_+$, $\{z \in \mathbb{R}_+^{L-1} : f(z) \geq q\}$ is convex, then z^f is convex-valued. If for each $q \in \mathbb{R}_+$, $\{z \in \mathbb{R}_+^{L-1} : f(z) \geq q\}$ is strictly convex, then z^f is single-valued.
6. (Sheppard's Lemma) Let $w \in \mathbb{R}_+^{L-1}$ and $q \in \mathbb{R}_+$. If $|z^f(w, q)| = 1$, then c^f is differentiable with respect to w at (w, q) and for each $l \in \{1, \dots, L-1\}$,

$$\frac{\partial c^f}{\partial w_l}(w, q) = z_l^f(w, q).$$

7. If z^f is a differentiable function in a neighborhood of $w \in \mathbb{R}_+^{L-1}$, then

$$(a) D_w z^f(w, q) w = 0,$$

$$(b) D_w z^f(w, q) \text{ is symmetric and negative semi-definite.}$$

8. If f is homogeneous of degree one, i.e., Y^f satisfies CR, then c^f and z^f are homogeneous of degree one in q .
9. If f is concave, then for each $w \in \mathbb{R}_+^{L-1}$, $c^f(w, \cdot)$ is a convex function.

4.5.1 Single-input and single-output model

In this section we study the single-input and single-output model. The simplicity of this environment allows us to graphically characterize a firm's cost minimization and profit maximization.

Consider a firm represented by the production function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$. Let $p \in \mathbb{R}_+$ be the output price and $w \in \mathbb{R}^+$ be the input price. In order to simplify notation we consider w to be fixed and denote $c^f(w, q)$ by $c(q)$. Likewise we denote $y^f(w, p)$ by $Q(p)$. The PMP is then

$$\max_{q \geq 0} pq - c(q).$$

Assume that this problem has a unique solution $Q(p)$. If this solution is interior and the cost function is differentiable, then

$$p = c'(Q(p)).$$

Definition 102. Let $q \in \mathbb{R}_+$. Then

- $AC(q) \equiv \frac{c(q)}{q}$ is the average cost at q .
- $MC(c) \equiv c'(q)$ is the marginal cost at q .

In what follows we explore how to graphically find c , AC , MC , and Q from f .

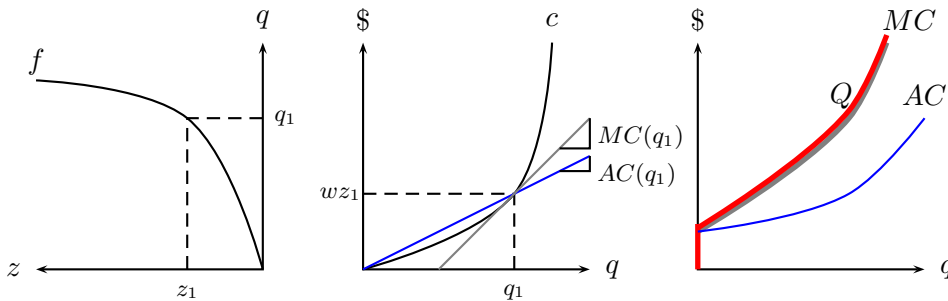


Figure 4.1: Finding c , AC , MC , and Q from f .

4.5.2 Long-run vs. short run

It is not standard what economists mean when they refer to production possibilities in the “long-run.” The most common interpretations are the following.

Long-run production possibilities under free-entry

Suppose that a firm is represented by a production set Y . If there is free entry of firms to this market, this firm can be replicated as many times as possible. Thus, in the “long-run” the aggregate technical restrictions are given by the set

$$AY \equiv \{y_1 + \cdots + y_n : n \in \mathbb{N}, \forall i = 1, \dots, n, y_i \in Y\}.$$

Proposition 55. If Y satisfies NI returns to scale, then AY satisfies CR to scale.

Under additional technical conditions one can guarantee that if Y is represented by a production function that is homogeneous of degree one. Thus, the supply correspondence is flat.

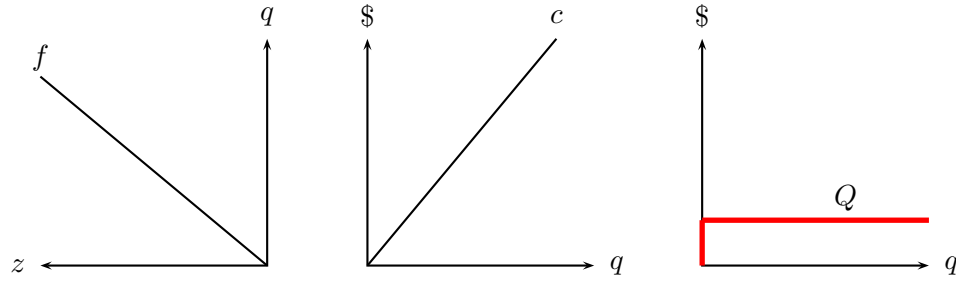


Figure 4.2: Finding c , AC , MC , and Q from f homogeneous of degree one.

Multiple input case with temporarily fixed inputs

In situations in which there are multiple inputs, some of these may be temporarily fixed. Economists refer to these situations as “short run.” Consistently, they refer to situations in which no input is fixed as the “long run.”

For instance, suppose that there are two inputs: labor, denoted by L , and capital, denoted by K . The “long run” production function is simply the production function as we have defined it in our production model

$$(L, K) \in \mathbb{R}_+^2 \mapsto f(L, K) \in \mathbb{R}_+.$$

For a fixed amount of capital, say $\bar{K}$, the short run production function is $f_{\bar{K}}$ defined by, for each L , $f_{\bar{K}}(L) \equiv f(L, \bar{K})$. The cost function associated

with $f_{\bar{K}}$ is the short run cost function at $\bar{K}$. One can easily graphically characterize the long run cost and average cost functions, as the lower envelope of short run cost and short run average cost functions. Clearly, under this interpretation, long run supply correspondences are not necessarily flat.

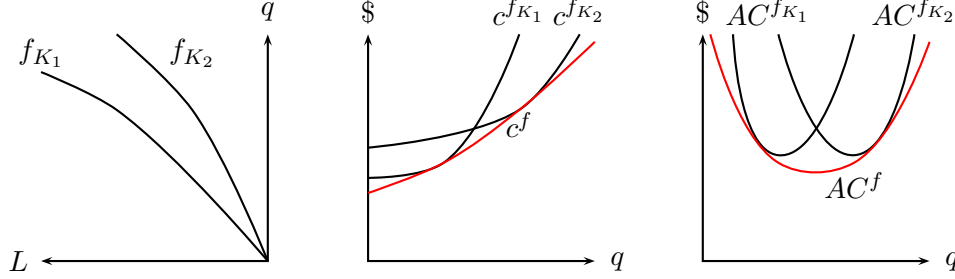


Figure 4.3: Relation of “short run” and “long run” cost functions.

4.6 Aggregate production

Let $\{Y_1, \dots, Y_J\}$ be a family of production sets. The aggregate production set associated with $\{Y_1, \dots, Y_J\}$ is the set

$$Y_{AG} \equiv Y_1 + \dots + Y_J \equiv \{y_1 + \dots + y_J : y_1 \in Y_1, \dots, y_J \in Y_J\}.$$

The aggregate supply correspondence associated with $\{Y_1, \dots, Y_J\}$ is the correspondence that associates with each $p \in \mathbb{R}^L$, the set

$$y_{AG}(p) \equiv y^{Y_1}(p) + \dots + y^{Y_J}(p) \equiv \{y_1 + \dots + y_J : y_1 \in y^{Y_1}, \dots, y_J \in y^{Y_J}\}.$$

Proposition 56. Let $\{Y_1, \dots, Y_J\}$ be a family of production sets. Then, for each $p \in \mathbb{R}^L$,

1. $y^{Y_{AG}}(p) = y_{AG}(p)$.
2. $\Pi^{Y_{AG}}(p) = \Pi^{Y_1}(p) + \dots + \Pi^{Y_J}(p)$.

Proof. Let $y \in y_{AG}(p)$. Then, $y = y_1 + \dots + y_J$ where $y_1 \in y^{Y_1}, \dots, y_J \in y^{Y_J}$. Thus, $y \in Y_{AG}$ and consequently, $p \cdot y \leq \Pi^{Y_{AG}}(p)$. Thus,

$$p \cdot y = \Pi^{Y_1}(p) + \dots + \Pi^{Y_J}(p) \leq \Pi^{Y_{AG}}(p).$$

We claim that $p \cdot y \geq \Pi^{Y_{AG}}(p)$. Suppose by contradiction that $p \cdot y < \Pi^{Y_{AG}}(p)$. Then, there is $y' \equiv y'_1 + \dots + y'_J \in Y_{AG}$ such that $p \cdot y < p \cdot y'$. Then, there

is $j \in \{1, \dots, J\}$ such that $p \cdot y'_j > p \cdot y_j$ for otherwise $p \cdot y'_1 + \dots + p \cdot y'_J \leq p \cdot y_1 + \dots + p \cdot y_J$. Thus, $y_j \notin y^{Y_j}(p)$. This is a contradiction.

Thus, $y \in y^{Y_{AG}}(p)$ and $p \cdot y = \Pi^{Y_{AG}}(p) = \Pi^{Y_1}(p) + \dots + \Pi^{Y_J}(p)$.

Now, let $y \in y^{Y_{AG}}$. Then, $p \cdot y = \Pi^{Y_{AG}}$ and there are $y_1 \in Y_1, \dots, y_J \in Y_J$ such that $y = y_1 + \dots + y_J$. Let $j \in \{1, \dots, J\}$. Then, $p \cdot y^j \leq \Pi^{Y_j}(p)$. We claim that $y_j \in y^{Y_j}(p)$. Suppose by contradiction that for some $j^* \in \{1, \dots, J\}$, $p \cdot y^{j^*} < \Pi^{Y_{j^*}}(p)$. Then, $p \cdot y^1 + \dots + p \cdot y^J < \Pi^{Y^1}(p) + \dots + \Pi^{Y^J}(p)$. Thus, $\Pi^{Y_{AG}}(p) < \Pi^{Y^1}(p) + \dots + \Pi^{Y^J}(p)$. This is a contradiction. $\square$

4.7 Efficient production

Definition 103. Let $Y \subseteq \mathbb{R}^L$ be a production set. A feasible vector $y \in Y$ is efficient if there is no $y' \in Y$ such that $y' \succeq y$.

Proposition 57. Let $Y \subseteq \mathbb{R}^L$ and $y \in Y$. Then,

1. If $y \in Y$ is profit maximizing at some $p \in \mathbb{R}_{++}^L$, then y is efficient.
2. Suppose that Y is convex. If $y \in Y$ is efficient for Y , then it is profit maximizing for some $p \geq 0$.

4.8 Problems

1. Prove that the converse of Statement 1 in Proposition 57 is not necessarily true. Prove that Statement 2 in the same proposition cannot be strengthened to require $p \gg 0$.